

AI Workflow Documentation

Supplementary Appendix to “Skin in the Chain: Locked Supply and the Cross-Section of Cryptocurrency

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Summary of AI Conversations (June 10 – August 10, 2026)

The project comprised 46 logged AI sessions across two environments: interactive Cowork conversations (Claude, Anthropic), where the author set direction, critiqued output, and made every substantive decision, and autonomous Claude Code build sessions, each launched from a written kickoff prompt fixed in advance and reviewed afterward under an honest-results clause forbidding specification changes after results were seen. Full verbatim transcripts are in Volume II; the repository (github.com/moazzamkhoja/AFA_2027_QTM_Crypto) archives the 46 contemporaneous session logs, 33 verbatim kickoff prompts, a 127-entry decisions log, and all code and data.

Phase A — Theory and literature (June 10–19; sessions 001–005, Cowork). The author shared the AFA call and a two-page QTM valuation framework. Conversations established the theoretical foundations: convenience yield (Sockin-Xiong) and monetary search (Kiyotaki-Wright) as primary frames, seigniorage (Cong et al.) as the income mechanism, the coin/token distinction, and the central construct — the conviction ratio λ , the locked fraction of supply, with the SoV/MoE decomposition following from QTM. The author rejected an early theta-coefficient formulation, demanded a rigorous continuum-of-agents aggregation to replace a flawed step, challenged an artificial model split, and approved the hypotheses (conviction level; valuation conditioning; quadrant portfolio). First drafts of the introduction and theory were written and revised through detailed author critique (eight tracked issues in one round).

Phase B — Data construction (June 22 – July 30; sessions 006–041, mostly Claude Code). Autonomous sessions built the empirical infrastructure from free, verifiable sources under standing constraints (no paid tiers; all joins on CoinMarketCap id; engine guards immutable): a 1,939-asset monthly universe (2015–2026); the three-channel conviction panel (staking from chain-native archives, HODL-6m from full transfer-log replay on thirteen EVM chains, governance delegation/voting); on-chain transaction-volume series validated within 5 percent of chain totals; the TVL panel; and the growth-levelized valuation ratios. The phase was iterative and adversarial with the data: fake archive nodes were detected and excluded, identity mismatches purged, contamination guards enforced, and every dead end (Monero’s hidden amounts, pruned nodes, paid-only vendors) logged rather than papered over. Final samples: 24 proof-of-stake coins (718 asset-months) and 101 tokens (2,771 asset-months).

Phase C — Empirical testing (August 4; sessions 042–045). A Cowork session fixed the analysis specification in advance (regression ladders, self-built monthly Liu-Tsyvinski-Wu factor analogs, portfolio rules); three Claude Code sessions then executed it. Results: the conviction level is rejected for coins but its interaction with valuation is strongly negative and survives a twelve-signal joint race; for tokens conditioning fails under six valuation measures, while conviction is priced through the extremes of its distribution — a quintile long-short earning 1.7 percent monthly factor-adjusted alpha that no competitor portfolio spans. Nulls (the quadrant portfolio, scarcity models) were reported at equal prominence. Mechanism tests (M1–M4) were postulated after the coin/token asymmetry emerged and tested once each; author-directed review analyses (within-sector quintiles, long-only implementation, channel decomposition) were run once and disclosed as exploratory.

Phase D — Writing, revision, and submission (August 4–10; Cowork). The author directed

successive pruning and rewriting rounds: hypotheses unified across asset classes, mechanisms moved to robustness with a postulate-then-test framing, tables restructured around the questions they answer, three introduction iterations (motivation-first opening; lambda as the bridge between QTM and tokenized models; findings led by the joint-race and spanning results), the final abstract in the author’s own words, and the title. The author caught and corrected several AI errors along the way, including a mischaracterization of practitioner valuation methods, a stale significance claim, and a misleading between-sector framing that a requested power analysis overturned. The phase closed with assembly of this documentation package.

Division of labor. The author contributed direction, theory judgment, all hypothesis and identification decisions, quality control, and final language; the AI contributed drafting, literature synthesis, all data engineering and estimation code, execution of pre-specified analyses, and revision mechanics. Every substantive claim in the paper traces to a decision recorded in the logs.

Line attribution (human vs. AI). Counted from the repository at submission: 17,502 lines of code (Python builders, analysis scripts, utilities), 2,019 lines of paper source (LaTeX), and 28,262 lines of documentation (session logs, decisions log, kickoff prompts, specifications, literature review) — 47,783 committed lines in total. Measured by who typed the committed line, the AI contributed over 99.9 percent: the author’s directly typed contributions are the final abstract text (drafted verbatim by the author) and scattered sentence-level wordings dictated in conversation. The author’s contribution is instead concentrated in the layer the line counts do not capture: the initial prompt and all subsequent direction, the theoretical judgments, every methodological decision (127 logged entries), quality control that caught and reversed AI errors, and approval of all final language — approximately 53 hours of logged human time against roughly two person-months of AI-executed construction, estimation, and writing.

Time Log by Conversation

Human time from the contemporaneous time log (66 entries; durations marked * are author estimates). AI session wall-clock time where recorded in session logs; autonomous sessions ran unattended.

Session	Date	Topic	Human (min)
001	2026-06-10	general	100
002	2026-06-10	theory	0
003	2026-06-12	general	105
004	2026-06-19	general	90
005	2026-06-22	general	125
006	2026-06-22	phase0 pipeline	0
007	2026-06-22	phase0 review	0
008	2026-06-22	phase0b followup	0
009	2026-06-23	phase1 lambda	35
010	2026-06-24	phase1 closeout phase2 pq	50*
011	2026-06-24	pq theory	0
012	2026-06-24	pq pilot	45*
013	2026-06-24	phase2 build	45*
014	2026-06-25	phase2b coins	55*
015	2026-06-25	dune pilot	10*
016	2026-06-25	dune dryrun	10*
017	2026-06-25	phase2c diagnostic	10*
018	2026-06-26	repo sync and lambda tvl scope	30
019	2026-06-26	lambda scale and tvl panel	10*
020	2026-06-26	bucket2 bucket3 recovery	30*
021	2026-06-29	token bucket1 exhaustive reaudit	10*
022	2026-06-29	etherscan nonEVM lambda channel audit	135*
023	2026-06-29	hex akro reconciliation survivorship	55*
024	2026-06-30	lambda free build	75*
025	2026-07-01	etherscan pro channel2 build	135*
026	2026-07-01	channel2 tail and op	75*
027		tvl expansion and ch2 tail	20*
028	2026-07-04	cowork coverage csv and session028 plan	45*
029		breadth build and coin probes	9*
030	2026-07-05	task a resume cut short	15*
031	2026-07-24	task a day1 complete	20*
032	2026-07-24	myx build task a closed	300*
033	2026-07-25	xtz matic nvt probes negative	0
034	2026-07-25	chz blockchair defi batch1	10*
035	2026-07-25	defi batch2	180*
036	2026-07-26	steth meme batch3a	60*
037	2026-07-27	dot ksm core ch1	60*
038	2026-07-27	shib ch2 batch3b	60*
039	2026-07-28	dotksm pq trx warp	60*
040	2026-07-28	cosmos lcd ch1	60*
041	2026-07-30	hxro sxp osmo	60*
042	2026-08-04	paper restructure tvlgl phase3 design	780*
043	2026-08-04	phase3 core tests	60*
044	2026-08-04	phase3b confirmatory sorts	60*
045	2026-08-04	phase3c fees dcf technicals	60*
046	2026-08-05	to 10 writing revision submission	0
Total human time			3154 min (~53 h)

AI compute: 36 autonomous Claude Code build sessions (typically 1–3 h wall-clock each where logged) plus interactive Cowork sessions concurrent with the human time above. Autonomous sessions required no human attention between launch and review.